A UNIT-COEFFICIENT d^{-3} MOTION BOUND FOR PRIMITIVE DISTANCE-REGULAR GRAPHS

MACHINE-ASSISTED PROOF DEVELOPMENT

Abstract. Let X be a primitive distance-regular graph on n vertices, of diameter $d \geq 3$. Subject to the published inputs identified precisely below, we prove that either X is a Johnson graph or a Hamming graph, or [LLM]

$$\mathrm{motion}(X) \geq \frac{n}{d^3}.$$

[LLM]

The unit coefficient comes from a Metsch-to-geometric parameter split: either Koolen–Bang applies directly, or integer line-counting activates Kivva's clique-geometry criterion. [Met99, Result 2.1; KB10, Theorem 5.3; K, Theorem 2.12 and Proposition 2.13 (arXiv numbering); LLM parameter split strengthening Proposition 4.1 of the preceding machine-assisted $2n/(3d^3)$ draft] The remaining new modules are a local-grid Johnson argument, exact retention of Kivva's dual-spectrum and incidence factors in the $\mu = 1$ branch, and an exact-envelope strengthening of Kivva Proposition 4.6 in the $\mu = 2$ branch. The two crosswalks below state the published baseline and exact change for each module, and the same pointers are repeated where the estimates are used. [Biggs71, Section 4, theorem on pp. 19–20; K, Theorem 1.2, Lemmas 2.27 and 5.5, Corollary 5.6, and Proposition 4.6; PS, Propositions 2.14 and 2.19; LLM summary of the explicitly mapped replacements and retained constants]

Verification status. Every imported statement is tagged by a theorem, result, equation, section, or page pointer; every new assertion is tagged [LLM]. The accompanying C++ checker and Python wrapper perform the finite and scalar checks with exact rational arithmetic. [LLM/editorial] The proof is machine-assisted research and is not a substitute for specialist peer review. In particular, the Metsch-to-geometric parameter split, local-grid rigidity argument, exact dual-spectrum transfer, and finite $\mu = 2$ certificate should be checked independently before the theorem is cited as established. [LLM/editorial]

Provenance convention. The label [PS, ...] refers to Pyber–Skresanov [1]; [BW, ...] to Babai–Wilmes [2]; and [K, ...] to Kivva [3]. The original-source labels are [Del73] for Delsarte [4], [Met95] and [Met99] for Metsch [6, 7], [KB10] for Koolen–Bang [5], [Biggs71] for Biggs [8], [T85] and [T86] for Terwilliger [9, 10], and [Ega81] for Egawa [11]. Journal numbering is used unless stated otherwise. [LLM/editorial]

[LLM] means that the statement is proved, defined, or algebraically derived here rather than copied from an identified source; it does not by itself assert conceptual novelty. A label of the form "[source; LLM retains/sharpens/replaces ...]" names both the published baseline and the exact change made here. Thus a source pointer is not intentionally omitted merely because the final formula is new. Statements marked only [LLM] are independent calculations for which no specific published baseline is being claimed. [LLM/editorial responding to source-audit request]

Original-source crosswalk. The exact roles of the older references are as follows; later annotations point back to these entries at the place of use. [LLM/editorial]

Date: July 24, 2026.

Label	Original pointer	Role in this proof
Del73	Section 3.3.2	Clique–coclique/Delsarte bound; Kivva Section 2.3 records the distance-regular form used here.
Met95/Met99	Met95 Result 2.2; Met99 Result 2.1	Parameterized line theorem; Kivva Theorem 2.12 and Koolen–Bang Proposition 5.1 are the exact modern restatements used here.
KB10	Theorem 5.3	Geometricity from $a_1 > M^2 c_2$ in a prescribed least-eigenvalue interval.
Biggs71	Section 4, theorem on pp. 19–20	Multiplicity formula; Kivva Theorem 2.10 gives the distance-regular standard-sequence form.
T85	Intersection-number inequalities arising from an induced quadrangle	Exact inequality is restated as Kivva Theorem 2.6.
T86	Main theorem	Local eigenvalue forced when an eigenvalue has multiplicity below k; exact form is Kivva Theorem 4.1.
Ega81	Main classification, pp. 108–125	Graphs with Hamming intersection numbers are Hamming or Doob; exact form is Kivva Theorem 2.24 (arXiv)/2.25 (journal).

Map of every claimed strengthening or replacement. This table records the baseline that each comparison label refers to. The same pointer is repeated locally where the strengthened estimate is used. [LLM/editorial responding to source-audit request]

New step	Baseline pointer	Exact comparison
Support-sensitive cut	PS Proposition 2.8, penultimate displayed inequality and the following weakening	Retains $(n - \|S\|)/n = 1 - \rho$; PS then replaces this factor by 1/2.
Geodesic spectral gap	PS Proposition 2.9	Replaces $k/(8D^2)$ by $n^2k/\sum_{x,y} \mathrm{dist}(x,y)^2 \geq k/D^2$.
Unit structural split	Proposition 4.1 of the preceding machine-assisted $2n/(3d^3)$ draft	Replaces that draft's coefficient-3/2 scalar split by the integer line-count split at $m = d - 1$; this is an internal comparison, not an attribution to a published source.
Johnson branch	K Theorem 1.2 and PS Proposition 2.19	Replaces the universal small-ε criterion by the local-grid rigidity argument in Proposition 6.1.
$\mu = 1$ branch	K Lemmas 2.27 and 5.5, followed by K Corollary 5.6 and PS Proposition 2.14	Retains the exact dual eigenvalue shift and the exact incidence factor $\beta/(\beta + 1)$ before the published simplifications to 1/2 and the stated $\eta/4$ motion bound.
$\mu = 2$ branch	Biggs71 Section 4; K Theorem 2.10 and Proposition 4.6	Uses the same Biggs denominator, replaces Kivva's $\varepsilon < 1/(6m^4d)$ loss envelope by exact rational envelopes at $\varepsilon = 2/(d^3 - 1)$, and on nineteen rows retains the additional t- or $(t-2)$-sphere summand.

Audit changes from the exploratory unit-coefficient draft. Two edge conditions are made explicit. Kivva's strict-growth lemma controls τ_i only through τ_{t-1}; every refined denominator below therefore uses $\psi_{t-1} \leq c_t$, not an unsupported inequality $\tau_t \geq t$. Also, the $m = 2$ contradiction uses Terwilliger's induced-quadrangle inequality at index $i = 2$, not at $i = 1$. [K, Lemma 4.2 and Theorem 2.6; T85, induced-quadrangle intersection-number inequalities; LLM correction of the exploratory draft]

1. Statement and parameters

Let X be a primitive distance-regular graph on n vertices, with valency k, diameter $d \geq 3$, intersection numbers b_i, c_i, and

$$\lambda = a_1, \qquad \mu = c_2.$$

[LLM/notation]

Write k_i for the size of a distance-i sphere, θ for the second largest adjacency eigenvalue, and $-m$ for the smallest eigenvalue, so $m > 0$. [LLM/notation]

Theorem 1.1. *Either X is a Johnson graph or a Hamming graph, or*

$$\text{motion}(X) \geq \frac{n}{d^3}. \tag{1}$$

[LLM]

Distance-regular graphs of valency 2 are cycles. [PS, Section 2.1] If $k = 2$, every nonidentity automorphism of a cycle moves at least $n - 2$ vertices, which is greater than n/d^3 for $d \geq 3$. [LLM] Henceforth assume

$$k > 2.$$

[PS, standing assumption after Section 2.1; LLM cycle dispatch]

Set

$$\gamma = \frac{1}{d^3}, \qquad \varepsilon = \frac{2}{d^3 - 1}. \tag{2}$$

[LLM/definition]

Then

$$\frac{\varepsilon(1-\gamma)}{2} = \gamma. \tag{3}$$

[LLM]

We shall repeatedly use

$$0 < \gamma < \frac{1}{2}, \qquad 0 < \varepsilon \leq \frac{1}{13} < \frac{1}{2}, \qquad \varepsilon < \frac{1}{d^2}. \tag{4}$$

[LLM]

The last two bounds are strongest at $d = 3$; the final one is equivalent to $d^3 - 2d^2 - 1 > 0$. [LLM]

2. Adjacent distinguishers and automorphism support

For vertices u, v, put

$$D(u, v) = \{z : \text{dist}(u, z) \neq \text{dist}(v, z)\}.$$

[LLM]

For adjacent u, v, the cardinality is constant by distance-regularity; denote it by $D(1)$. [LLM/notation]

Lemma 2.1 (Exact adjacent-pair identity)**.** *For every distance-regular graph of diameter at least three,* [LLM]

$$D(1) = 2 + \frac{2}{k} \sum_{i=2}^{d} k_i c_i. \tag{5}$$

[LLM]

Consequently, [LLM]

$$D(1) > \frac{\mu}{k} n. \tag{6}$$

[LLM]

Proof. For adjacent u, v, the number of vertices at distance i from both is $p^1_{i,i}$. [LLM/notation] A double count gives the balance identity

$$k p^1_{i,i} = k_i p^i_{1,i} = k_i a_i,$$

[LLM, double count]

so

$$D(1) = n - \frac{1}{k} \sum_{i=1}^{d} k_i a_i.$$

[LLM]

Using $a_i = k - b_i - c_i$ and the standard sphere recurrence $k_i b_i = k_{i+1} c_{i+1}$, together with $b_d = 0$ and $k_1 c_1 = k$, we obtain [PS, Section 2.1; K, Section 2.2, (3′)]

$$\sum_{i=1}^{d} k_i a_i = k(n-2) - 2 \sum_{i=2}^{d} k_i c_i,$$

[LLM, telescoping]

which proves (5). [LLM]

We also use the comparison $c_i \leq b_j$ whenever $i + j \leq d$. Place vertices u, w, v in this order on a geodesic with $\text{dist}(u, w) = i$ and $\text{dist}(w, v) = j$. Every neighbor z of w with $\text{dist}(u, z) = i - 1$ must satisfy $\text{dist}(v, z) = j + 1$, so the c_i such neighbors form a subset of the b_j such neighbors. [LLM] In particular $c_2 \leq b_1$, hence

$$k_2 = \frac{k b_1}{c_2} \geq k_1 = k.$$

[K, Section 2.2, (3′); LLM]

It follows that $\sum_{i=2}^{d} k_i \geq (n-1)/2$. [LLM] Since $c_i \geq c_2 = \mu$ for $i \geq 2$, [PS, Section 2.1]

$$D(1) \geq 2 + \frac{\mu}{k}(n-1) > \frac{\mu}{k}n,$$

[LLM]

where the final inequality uses $\mu \leq k$. [PS, Section 2.1; LLM] □

Lemma 2.2 (Support-sensitive adjacent pair). *Let $g \in \mathrm{Aut}(X)$ be nonidentity, put $S = \mathrm{supp}(g)$ and $\rho = |S|/n \leq 1/2$. Some $x \in S$ has at least* [PS, Proposition 2.8, penultimate inequality; LLM retains PS's factor $(n-|S|)/n = 1-\rho$ before PS weakens it to $1/2$]

$$\frac{k}{d}(1-\rho) \tag{7}$$

[PS, Proposition 2.8, penultimate inequality; LLM rewrites $(n-|S|)/n$ as $1-\rho$]

fixed neighbors. If $\mu < k(1-\rho)/d$, then $x \sim x^g$ and [LLM]

$$\lambda \geq \frac{k}{d}(1-\rho), \qquad D(1) \leq |S|. \tag{8}$$

[LLM]

Proof. We spell out the orientation convention while retaining the sharper penultimate bound in Pyber–Skresanov Proposition 2.8, rather than its final replacement of $(n-|S|)/n$ by $1/2$. [PS, Proposition 2.8, especially the penultimate displayed inequality; LLM orientation bookkeeping only] For a directed edge e of X, let

$$P_e = \sum_{u,v} \frac{1}{p(u,v)} \#\{L : L \text{ is a } u\text{–}v \text{ geodesic using } e\},$$

[PS, proof of Proposition 2.8]

where $p(u,v)$ is the number of u–v geodesics. [PS, proof of Proposition 2.8] Their coherent-configuration argument shows that P_e is independent of the directed edge; write the common value as P. [PS, Lemma 2.7 and proof of Proposition 2.8] Summing over the nk directed edges gives

$$nkP = \sum_{u,v} \mathrm{dist}(u,v) \leq dn^2, \qquad \text{so} \qquad P \leq \frac{dn}{k}.$$

[PS, proof of Proposition 2.8]

Let

$$\delta_X^+(S) = \{(u,v) : u \in S,\ v \notin S,\ u \sim v\}.$$

[LLM/notation]

For each ordered pair $(x,y) \in S \times (V(X) \setminus S)$, every x–y geodesic uses at least one edge of $\delta_X^+(S)$. [PS, proof of Proposition 2.8] Consequently

$$|S|(n-|S|) \leq |\delta_X^+(S)|P,$$

[PS, proof of Proposition 2.8]

and hence [LLM]

$$|\delta_X^+(S)| \geq |S|\frac{k}{d}\frac{n-|S|}{n}. \tag{9}$$

[PS, Proposition 2.8, penultimate displayed inequality; LLM directed-edge notation]

Each undirected cut edge contributes exactly one directed edge to $\delta_X^+(S)$, so this is also the corresponding undirected cut size. Averaging the outbound cut degrees over the vertices of S proves (7). [LLM, directed/undirected conversion] Since S is the support of g, every neighbor outside S is fixed. [LLM]

Every fixed neighbor of x is also adjacent to x^g. If x and x^g were nonadjacent, the existence of a common neighbor would put them at distance two, where they have exactly μ common neighbors. This contradicts the strict hypothesis. Thus $x \sim x^g$, and the fixed common neighbors give the lower bound on λ. [LLM]

For $z \notin S$,

$$\mathrm{dist}(x,z) = \mathrm{dist}(x^g, z^g) = \mathrm{dist}(x^g, z),$$

[LLM]

so no fixed vertex distinguishes x and x^g. Hence $D(1) \leq |S|$. [LLM] □

3. A direct geodesic Poincaré inequality

The next proposition is the self-contained spectral improvement used in the $\mu = 1$ branch. [LLM]

Proposition 3.1 (Geodesic spectral-gap bound)**.** *Let G be the graph of a connected symmetric basis relation of valency k in a homogeneous coherent configuration on n points. If D is its diameter and θ_1 its second largest adjacency eigenvalue, then* [LLM]

$$k - \theta_1 \geq \frac{n^2 k}{\sum_{x,y \in V(G)} \operatorname{dist}(x,y)^2} \geq \frac{k}{D^2}. \tag{10}$$

[LLM]

For a distance-regular graph, [LLM]

$$k - \theta_1 \geq \frac{nk}{\sum_{i=0}^{D} k_i i^2}. \tag{11}$$

[LLM]

Proof. For ordered vertices x, y, let $p(x,y)$ be the number of geodesics from x to y. For a directed edge e, define [PS, proof of Proposition 2.8, unweighted load P_e; LLM inserts the relation-constant weight $\operatorname{dist}(x,y)$]

$$Q_e = \sum_{x,y} \frac{\operatorname{dist}(x,y)}{p(x,y)} \#\{P : P \text{ is an } x\text{–}y \text{ geodesic containing } e\}.$$

[PS, proof of Proposition 2.8, definition of P_e; LLM weighted analogue with one extra factor $\operatorname{dist}(x,y)$]

The intersection-number argument in Pyber–Skresanov's proof of Proposition 2.8 shows that Q_e is independent of e. After fixing the three basis relations containing (x,z), (w,y), and (x,y) for $e = (z,w)$, the number of relevant pairs and geodesics is independent of (z,w) by their Lemma 2.7; the extra weight $\operatorname{dist}(x,y)/p(x,y)$ depends only on the basis relation containing (x,y). Write the common value as Q. [PS, Lemma 2.7 and proof of Proposition 2.8; LLM observes that the added weight is constant on the relation containing (x,y)]

Summing over the nk directed edges gives

$$nkQ = \sum_{x,y} \operatorname{dist}(x,y)^2, \tag{12}$$

[LLM, weighted load double count]

because every geodesic of length ℓ contributes ℓ at each of its ℓ directed edges. [LLM]

Let f have mean zero. Averaging Cauchy–Schwarz along all geodesics from x to y gives [LLM]

$$(f(x) - f(y))^2 \leq \frac{\operatorname{dist}(x,y)}{p(x,y)} \sum_{P : x \to y} \sum_{e \in P} (\nabla_e f)^2.$$

[LLM]

After summing over ordered pairs and using the constancy of Q_e, [PS, Lemma 2.7; LLM]

$$2n\|f\|_2^2 \leq Q \sum_{e \text{ directed}} (\nabla_e f)^2 = 2Q f^{\mathsf{T}}(kI - A)f.$$

[LLM]

Taking f in the θ_1-eigenspace gives $k - \theta_1 \geq n/Q$. Equation (12) proves the first inequality in (10); the bound $\operatorname{dist}(x,y) \leq D$ proves the second. [LLM] In a distance-regular graph,

$$\sum_{x,y} \operatorname{dist}(x,y)^2 = n \sum_{i=0}^{D} k_i i^2,$$

[LLM]

which proves (11). [LLM] □

Remark 3.2. Pyber and Skresanov obtain $k - \theta_1 \geq k/(8D^2)$ by passing from their Proposition 2.8 geodesic-expansion estimate through Cheeger's inequality. [PS, Propositions 2.8–2.9] Proposition 3.1 applies Cauchy–Schwarz directly to the same uniform loads and proves $k - \theta_1 \geq n^2 k / \sum_{x,y} \operatorname{dist}(x,y)^2 \geq k/D^2$, improving the displayed PS constant by a factor of eight. [PS, Proposition 2.9; LLM direct-energy strengthening] No novelty claim is made for this canonical-path inequality without a broader literature search. [LLM/editorial]

4. Small support forces Delsarte geometry

We use Metsch's theorem in the following exact specialization. If a connected graph has exactly λ common neighbors for adjacent pairs, at most μ for nonadjacent pairs, and an integer $s \geq 1$ satisfies

$$\lambda > (2s-1)(\mu-1)-1, \qquad k < (s+1)(\lambda+1) - \frac{s(s+1)}{2}(\mu-1), \tag{13}$$

[Met99, Result 2.1; Met95, Result 2.2; KB10, Proposition 5.1; K, Theorem 2.12 (arXiv); BW, Theorem 1.2]

then the maximal cliques of size at least

$$\lambda + 2 - (s-1)(\mu-1)$$

[Met99, Result 2.1; Met95, Result 2.2; K, Theorem 2.12 (arXiv)]

form a clique geometry, and every vertex lies on at most s of these cliques. [Met99, Result 2.1; Met95, Result 2.2; KB10, Proposition 5.1; K, Theorem 2.12 (arXiv)]

Proposition 4.1 (Metsch-to-geometric structural reduction)**.** *Suppose that a nonidentity automorphism has support density $\rho < \gamma$. Then*

$$\mu < \gamma k, \qquad \lambda > \frac{1-\gamma}{d}k, \qquad k > d^3, \tag{14}$$

[LLM]

and X is Delsarte-geometric with smallest eigenvalue $-m$ satisfying

$$m \leq d. \tag{15}$$

[LLM]

Proof. Put

$$\alpha = \frac{1-\gamma}{d}.$$

[LLM/definition]

Suppose first that $\mu \geq \alpha k$. Let $k_i = k_{\max}$ be a largest nontrivial relation valency. Since $k_2 \geq k_1$ by Lemma 2.1, we may take $i \geq 2$. [LLM] Monotonicity of the c_i and the sphere recurrence give

$$\frac{k_i}{k_{i-1}} = \frac{b_{i-1}}{c_i} \leq \frac{k}{\mu} \leq \frac{1}{\alpha},$$

[PS, Section 2.1 sphere recurrence and monotonicity; LLM]

so $n - k_{\max} \geq k_{i-1} \geq \alpha k_{\max}$. [LLM] Every nontrivial relation of the primitive distance scheme is connected and has diameter at most d; hence

$$\operatorname{motion}(X) \geq \frac{n-k_{\max}}{d}.$$

[PS, paragraph before Lemma 2.7 and Propositions 2.10, 2.12]

If $k_{\max} \geq n/2$, this is at least $\alpha n/(2d) > \gamma n$ because $d^3 - 2d^2 - 1 > 0$. If $k_{\max} < n/2$, it is greater than $n/(2d) > \gamma n$. Both alternatives contradict $\rho < \gamma$. [LLM] Therefore

$$\mu < \alpha k. \tag{16}$$

[LLM]

Since $\rho < \gamma < 1/2$, equation (16) implies

$$\mu < \frac{1-\rho}{d}k.$$

[LLM]

Lemma 2.2 therefore applies and gives

$$\lambda > \alpha k.$$

[LLM]

Together with Lemma 2.1 and $D(1) \leq |S|$, it gives

$$\mu < \rho k < \gamma k.$$

[LLM]

Since $\mu \geq 1$, we also have $k > 1/\gamma = d^3$. [LLM]
For $d \geq 3$,

$$\alpha > (2d-1)\gamma, \qquad (d+1)\alpha - \frac{d(d+1)}{2}\gamma > 1. \tag{17}$$

[LLM]

After multiplication by the positive denominators, the two numerators are

$$d^3 - 2d^2 + d - 1, \qquad d^3 - d^2 - 2d - 2,$$

[LLM]

and after writing $d = x + 3$ they become

$$x^3 + 7x^2 + 16x + 11, \qquad x^3 + 8x^2 + 19x + 10.$$

[LLM exact polynomial certificates]

Thus Metsch's theorem applies with $s = d$. Let $\mathcal{C}$ be the resulting special-line geometry. Every vertex belongs to at most d lines, and every edge lies in a unique line. [Met99, Result 2.1; KB10, Proposition 5.1; K, Theorem 2.12 (arXiv); LLM verifies the two hypotheses at (17)]

Put

$$f = \alpha - (d-1)\gamma = \frac{(d-1)(d^2+1)}{d^4} > 0.$$

[LLM]

Every line $C \in \mathcal{C}$ satisfies

$$|C| - 1 \geq \lambda + 1 - (d-1)(\mu - 1) > fk + d > d. \tag{18}$$

[Met99, Result 2.1; K, Theorem 2.12 (arXiv); LLM substitutes $s = d$ and the strict support bounds]

Let q_v be the number of lines in $\mathcal{C}$ containing v. Since the lines partition the edges incident with v,

$$k = \sum_{\substack{C \in \mathcal{C} \\ v \in C}} (|C| - 1).$$

[K, definition of clique geometry; LLM]

The Delsarte clique bound gives $|C| - 1 \leq k/m$ for every line, so

$$q_v \geq m.$$

[Del73, Section 3.3.2, clique–coclique bound; K, Section 2.3, distance-regular form; LLM sums the line sizes at v]

Because $q_v \leq d$, this proves $m \leq d$. [LLM]

Suppose first that $m \leq d - 1$, and put $M = \lceil m \rceil \leq d - 1$. Then

$$-M \leq -m < 1 - M, \qquad M^2\mu \leq (d-1)^2\gamma k < \alpha k < \lambda.$$

[LLM; the strict scalar inequality is equivalent to $2d^2 - d - 1 > 0$]

Koolen–Bang Theorem 5.3 applies with the integer parameter M, so X is Delsarte-geometric (and in particular its smallest eigenvalue is $-M$). [KB10, Theorem 5.3; PS, Proposition 2.5 is the reformulation used in the preceding version; LLM supplies $M = \lceil m \rceil$ and checks the eigenvalue interval]

It remains to consider $d - 1 < m \leq d$. Since q_v is an integer and $m \leq q_v \leq d$, one has $q_v = d$ for every vertex. [LLM integer rounding] The special lines form a clique geometry, every vertex lies on exactly d lines, and

$$k > d^3 \geq d^2.$$

[LLM]

Kivva's clique-geometry criterion therefore applies with its integer parameter equal to d: X is geometric and its smallest eigenvalue is $-d$. [K, Proposition 2.13 (arXiv numbering); LLM hypothesis check] Since the actual smallest eigenvalue was denoted by $-m$, this also gives $m = d$. [LLM] This proves the proposition. [LLM] ☐

5. The transition and spectral branches

Assume from now on that a nonidentity automorphism has support density $\rho < \gamma$. Proposition 4.1 is in force. [LLM]

Lemma 5.1 (Transition without a diameter loss)**.** *If*

$$b_j \geq \varepsilon k, \qquad c_{j+1} \geq \varepsilon k$$

[LLM]

for some $1 \leq j \leq d - 1$*, then* $|S| > \varepsilon n > \gamma n$*.* [LLM]

Proof. For $i \leq j$, monotonicity gives $b_i \geq \varepsilon k$; for $i \geq j + 1$, it gives $c_i \geq \varepsilon k$. [PS, Section 2.1] Thus $a_i \leq (1 - \varepsilon)k$ for every $i \geq 1$, and

$$D(1) = n - \frac{1}{k}\sum_{i=1}^{d} k_i a_i \geq n - (1 - \varepsilon)(n - 1) > \varepsilon n.$$

[LLM]

Now use $D(1) \leq |S|$ from Lemma 2.2; also $\varepsilon > \gamma$ follows from (2). [LLM] ☐

If no transition occurs, let t be the least index with $b_t \le \varepsilon k$, using $b_d = 0$. [LLM] The inequality $2\lambda \le k + \mu$ gives

$$b_1 = k - \lambda - 1 \ge \frac{k - \mu - 2}{2} > \frac{1 - 3\gamma}{2} k > \frac{k}{3},$$

[PS, Lemma 2.4; LLM]

where $1/k < \gamma$ was used. Hence $t \ge 2$. Since $b_{t-1} > \varepsilon k$ and no transition occurs,

$$b_t \le \varepsilon k, \qquad c_t < \varepsilon k. \tag{19}$$

[LLM]

Lemma 5.2 (High second eigenvalue)**.** *Under these assumptions,*

$$\theta \ge (1 - \varepsilon) b_1. \tag{20}$$

[LLM]

Proof. Suppose instead that $\theta < (1-\varepsilon)b_1$. Since $m \le d$, $b_1 > k/3$, $k > d^3$, and $\varepsilon < 1/2$,

$$(1-\varepsilon)b_1 > \frac{k}{6} > \frac{d^3}{6} > d \ge m.$$

[LLM]

Thus the zero-weight spectral radius $\xi = \max\{\theta, m\}$ satisfies $\xi < (1-\varepsilon)b_1$. [PS, notation before Proposition 2.13; LLM]

Moreover $\lambda > \mu$, because $(1-\gamma)/d > \gamma$. Two distinct vertices have λ, μ, or 0 common neighbors according as their distance is 1, 2, or at least 3; the common-neighbor maximum is therefore λ. [LLM] Babai's spectral motion bound gives

$$\begin{aligned} \rho &\ge \frac{k - \xi - \lambda}{k} > \frac{1 + \varepsilon b_1}{k} \\ &= \frac{1-\varepsilon}{k} + \varepsilon \frac{k - \lambda}{k} \ge \frac{1-\varepsilon}{k} + \frac{\varepsilon}{2}\left(1 - \frac{\mu}{k}\right) \\ &> \frac{\varepsilon}{2}(1-\gamma) = \gamma, \end{aligned}$$

[PS, Proposition 2.13; LLM]

contrary to $\rho < \gamma$. The final equality is (3). [LLM] □

6. The Johnson and $\mu = 1$ endgames

Proposition 6.1 (The case $\mu \ge 3$)**.** *Under* (19) *and* (20)*, if $\mu \ge 3$, then X is a Johnson graph.* [K, Theorems 3.1, 3.9, 2.21, Lemma 3.12 and Proposition 3.11; LLM local-grid proof replacing the universal small-ε criterion in K, Theorem 1.2 and PS, Proposition 2.19]

Proof. If a neighborhood graph is disconnected, Kivva's Proposition 3.11 gives

$$\theta + 1 \le \frac{5}{7} b_1,$$

[K, Proposition 3.11]

contradicting (20), because $1 - \varepsilon \ge 12/13 > 5/7$. Thus all neighborhood graphs are connected. [K, Lemma 2.20; LLM]

Put $\beta = k/m$. In a Delsarte geometry, m and β are positive integers. [K, Lemma 2.14] Fix v. The m Delsarte cliques through v partition $X(v)$ into parts $P_1, \dots, P_m$, each of size β. [K, Lemma 2.14 and proof of Lemma 3.8] Put

$$r = \psi_1 - 1.$$

[LLM/definition]

A vertex of P_i has r neighbors in every P_j with $j \ne i$; consequently each bipartite graph between two parts is r-regular. Connectedness gives $r \ge 1$. [K, definition of ψ_1, Lemma 2.20 and proof of Lemma 3.8]

Let

$$b^+ = \frac{b_1}{\theta + 1}.$$

[K, Theorem 3.1; LLM notation]

By (20),

$$b^+ < \frac{1}{1-\varepsilon} \le \frac{13}{12},$$

[LLM]

and Terwilliger's local bound gives

$$\lambda_{\min}(X(v)) \ge -1 - b^+.$$

[K, Theorem 3.1]

We claim $r = 1$. Suppose $r \ge 2$, and let B be the $\beta \times \beta$ bipartite incidence matrix between two parts. The restriction of $BB^\top$ to $\mathbf{1}^\perp$ has trace $\beta r - r^2$, so a nonprincipal singular value σ satisfies

$$\sigma^2 \ge \frac{r(\beta - r)}{\beta - 1}.$$

[LLM, trace averaging]

Kivva's inequality $\tau_2 \geq \psi_1$ gives $\mu = \tau_2\psi_1 \geq (r+1)^2$. Since $\mu < \gamma m\beta$ and $m \leq d$,

$$\beta > d^2(r+1)^2 > 6r.$$

[K, Lemmas 2.17–2.18; LLM]

Hence

$$\sigma^2 > \frac{5r}{6} \geq \frac{5}{3} > \left(\frac{13}{12}\right)^2 > (b^+)^2.$$

[LLM]

On vectors orthogonal to the all-ones vector in both parts, the induced two-part adjacency matrix has eigenvalue $-1-\sigma$. Interlacing contradicts the local lower bound. [LLM, singular-vector construction and interlacing] Thus $r = 1$, equivalently $\psi_1 = 2$. [LLM]

Every bipartite graph between two parts is now a perfect matching. If $m = 2$, this already gives $X(v) \cong K_2 \square K_\beta$. [LLM] Assume $m \geq 3$ and label each P_i by its matching with P_1. If a matching between P_i and P_j is represented by a nonidentity permutation, choose a cycle of length $\ell \geq 2$ and a nontrivial ℓth root of unity ζ with $\operatorname{Re}\zeta \leq -1/2$. [LLM] On the corresponding 3ℓ vertices, the Fourier mode reduces the adjacency matrix to $-I_3 + M_\zeta$, where

$$M_\zeta = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & \zeta \\ 1 & \overline{\zeta} & 0 \end{pmatrix}, \qquad \det(xI - M_\zeta) = x^3 - 3x - 2\operatorname{Re}\zeta.$$

[LLM, Fourier decomposition]

At $x = -3/2$ this polynomial is positive and it tends to $-\infty$ as $x \to -\infty$. Thus the induced local subgraph has an eigenvalue below $-5/2$, contradicting

$$\lambda_{\min}(X(v)) \geq -1 - b^+ > -\frac{25}{12} > -\frac{5}{2}.$$

[K, Theorem 3.1; LLM]

Therefore the matchings are globally consistent, and for every vertex

$$X(v) \cong K_m \square K_\beta = L(K_{m,\beta}).$$

[LLM]

Since $K_{m,\beta}$ is triangle-free, Kivva's Lemma 3.12 supplies an induced $K_{\tau_2,2}$. [K, Lemma 3.12] If $\tau_2 \geq 3$, Theorem 3.9 applied to an induced $K_{3,2}$ gives $b^+ \geq 7/5$, contrary to $b^+ < 13/12$. [K, Theorem 3.9; LLM] Hence $\tau_2 = 2$ and

$$\mu = \tau_2\psi_1 = 4.$$

[K, Lemma 2.17; LLM]

Kivva's local characterization makes X a Johnson graph or a quotient doubly covered by $J(2d, d)$. [K, Theorem 2.21] The exceptional alternative is excluded in Kivva's proof when $k \geq m^3$; here $k > d^3 \geq m^3$. [K, proof of Theorem 1.2 after Lemma 3.12; LLM hypothesis check] Thus X is Johnson. [LLM] □

Proposition 6.2 (The case $\mu = 1$)**.** *Under Proposition 4.1, the case $\mu = 1$ contradicts $\rho < \gamma$.* [K, Lemmas 2.26–2.27, 5.1 and 5.5, Proposition 5.13; PS, Proposition 2.14; LLM retains Kivva's exact dual eigenvalue shift and Lemma 5.5 incidence factor instead of the later $1/2$ and $\eta/4$ relaxations]

Proof. Proposition 3.1 gives

$$\theta \leq k\left(1 - \frac{1}{d^2}\right).$$

[LLM]

Put $\eta = 1/d^2$ and $\beta = k/m$. Since $\mu = \tau_2\psi_1 = 1$, integrality gives $\tau_2 = \psi_1 = 1$. [K, Lemmas 2.17–2.18; LLM]

Assume $m \geq 3$, and let $\widetilde{X}$ be the dual graph. Kivva's Lemma 2.26 gives

$$\widetilde{k} = (m-1)(\beta+1), \qquad q(\widetilde{X}) = m - 2,$$

[K, Lemma 2.26 and discussion before Theorem 5.2]

where q is the maximum number of common neighbors of distinct vertices. Since $k > d^3 \geq m^2$, Lemma 2.27 applies. [K, Lemma 2.27; LLM hypothesis check] Its exact spectral shift and the preceding gap give

$$\widetilde{\xi} \leq \widetilde{k} - \eta k.$$

[K, Lemma 2.27, exact shift $\widetilde{\theta} = \theta - k/m + m - 1$; LLM combines it with Proposition 3.1's gap k/d^2, rather than PS Proposition 2.14's abstract η input]

The positive nonprincipal eigenvalues satisfy this bound directly. The negative side has absolute value at most $\beta + 1$, and

$$(m-2)(\beta+1) \geq \eta m\beta$$

[LLM]

shows $\beta + 1 \le \widetilde{k} - \eta k$. [LLM]

Babai's spectral motion lemma therefore yields

$$\frac{\mathrm{motion}(\widetilde{X})}{|V(\widetilde{X})|} \ge \frac{\eta k - (m-2)}{\widetilde{k}}.$$

[K, Lemma 5.1; LLM substitution]

Kivva's incidence-transfer Lemma 5.5 gives the exact moved-point factor $(s-1)/s$ with $s = \beta + 1$, namely $\beta/(\beta+1)$; Kivva Corollary 5.6 subsequently weakens this to $1/2$. Retaining the lemma's exact factor gives

$$\frac{\mathrm{motion}(X)}{n} \ge \frac{\beta\big(m\beta/d^2 - m + 2\big)}{(m-1)(\beta+1)^2}. \tag{21}$$

[K, Lemma 5.5, exact factor $(s-1)/s$; K, Corollary 5.6, later $1/2$ relaxation; LLM substitutes $s = \beta + 1$ without that relaxation]

Because $m\beta = k > d^3$,

$$\frac{m\beta}{d^2} - m + 2 = \frac{m\beta}{d^2}\left(1 - \frac{(m-2)d^2}{m\beta}\right) > \frac{2m\beta}{d^3}.$$

[LLM]

Also $\beta > d^3/m \ge d^2 \ge 9$. Therefore (21) gives

$$\frac{\mathrm{motion}(X)}{n} > \frac{2m}{m-1}\left(\frac{\beta}{\beta+1}\right)^2 \frac{1}{d^3} > \frac{81}{50d^3} > \gamma.$$

[LLM]

If $m < 3$, geometric integrality gives $m = 1$ or 2. [K, Lemma 2.14] The case $m = 1$ forces X to be complete, contrary to $d \ge 3$. [LLM] Thus $m = 2$. Since $k > d^3 > 4$, Kivva's Proposition 5.13 gives

$$\mathrm{motion}(X) \ge \frac{n}{16} > \frac{n}{d^3}.$$

[K, Proposition 5.13; LLM hypothesis check]

□

7. The unit-coefficient Hamming-stability lemma

The remaining branch is $\mu = 2$. Kivva Proposition 4.6 obtains $f_1 < k$ from a single retained term $k_{t-1}u_{t-1}^2$ in Biggs' denominator under $\varepsilon < 1/(6m^4 d)$. The proof below preserves that one-term strategy generically, replaces its coarse loss estimates by exact rational envelopes at $\varepsilon = 2/(d^3-1)$, and on nineteen rows also keeps the t- or $(t-2)$-sphere term. [Biggs71, Section 4, theorem on pp. 19–20; K, Theorem 2.10 and Proposition 4.6; LLM strengthens K, Proposition 4.6 by replacing its $\varepsilon < 1/(6m^4 d)$ loss envelope and, on the listed rows, retaining additional nonnegative Biggs-denominator terms]

Proposition 7.1 (The case $\mu = 2$)**.** *Let X be Delsarte-geometric of diameter $d \ge 3$, with smallest eigenvalue $-m$, where $m \le d$, and suppose $\mu = 2$. If, for some $2 \le t \le d$,*

$$b_t \le \varepsilon k, \qquad c_t < \varepsilon k, \qquad \theta \ge (1-\varepsilon)b_1, \qquad k > d^3, \tag{22}$$

[LLM]

then X is a Hamming graph. [LLM]

Proof. Put $\beta = k/m$. Kivva's geometric identities are

$$c_i = \tau_i \psi_{i-1}, \qquad b_i = (m - \tau_i)(\beta + 1 - \psi_i). \tag{23}$$

[K, Lemma 2.17 (journal); Lemma 2.16 (arXiv)]

Here $\tau_1 = 1$. Kivva's inequality $\tau_2 \ge \psi_1$ and $\mu = 2$ give

$$\tau_2 = 2, \qquad \psi_1 = 1, \qquad b_1 = (m-1)\beta, \qquad \lambda = \beta - 1. \tag{24}$$

[K, Lemmas 2.17–2.18 (journal); LLM integrality]

Every neighborhood graph is consequently a disjoint union of m cliques. [K, Lemma 2.20 (journal)] Since $\mu = 2$, X contains an induced quadrangle, so the Terwilliger inequality

$$b_{i-1} \ge b_i + c_{i-1} - c_i + \lambda + 2 \qquad (1 \le i \le d)$$

[T85, induced-quadrangle intersection-number inequalities; K, Theorem 2.6, exact restatement]

is available wherever cited below. [K, Lemma 3.10 and Theorem 2.6; T85]

Because $\varepsilon < 1/d^2 \le 1/m^2$, Kivva's Lemma 4.2 gives

$$\tau_i < \tau_{i+1} \qquad (1 \le i \le t-2). \tag{25}$$

[K, Lemma 4.2; LLM hypothesis check]

Thus $\tau_i \ge i$ for $i \le t-1$. Put

$$r = m - \tau_{t-1}.$$

[LLM/definition]

Since $b_{t-1} > 0$,

$$1 \le r \le m - t + 1, \qquad 2 \le t \le m. \tag{26}$$

[K, Lemma 2.17; LLM]

If $m = 2$, then $t = 2 < d$. Terwilliger's inequality at $i = 2$ gives

$$b_1 \ge b_2 + \lambda + 2 + c_1 - c_2 = b_2 + b_1,$$

[T85, induced-quadrangle intersection-number inequality; K, Theorem 2.6, exact restatement; LLM corrects the exploratory draft's use of $i = 1$ to the valid index $i = 2$]

impossible because $b_2 > 0$. Hence $m \ge 3$. [LLM]

Let $(u_i)_{i=0}^d$ be the standard sequence for θ, and whenever $u_{i-1} > 0$ put

$$y_i = \frac{u_{i-1} - u_i}{u_{i-1}}.$$

[LLM/definition]

The standard-sequence recurrence becomes

$$y_{i+1} = \frac{k - \theta + c_i y_i/(1 - y_i)}{b_i}. \tag{27}$$

[K, Definition 2.9 and recurrence; LLM change of variables]

Set

$$B = 1 + (m-1)\varepsilon, \qquad u_* = (1 - \varepsilon)\frac{m-1}{m}. \tag{28}$$

[LLM/definition]

Then

$$k - \theta \le \beta B, \qquad u_1 = \frac{\theta}{k} \ge u_*.$$

[K, Definition 2.9; LLM]

We record the possible lower bounds on c_t. Monotonicity and (25) give

$$c_t \ge c_{t-1} = \tau_{t-1}\psi_{t-2} \ge t - 1. \tag{29}$$

[PS, Section 2.1; K, Lemmas 2.17 and 4.2; LLM]

If $c_t = t - 1$, equality forces

$$4 \le t \le m - 1, \qquad r = m - t + 1. \tag{30}$$

[K, Corollary 2.8 and Theorem 2.6; T85, induced-quadrangle inequality; LLM equality analysis]

If $t = m < d$, then $r = 1$, $c_{m-1} \ge m - 1$, and $b_m \ge 1$. Terwilliger's inequality at index m excludes $c_m = m$. If $c_m = m + 1$, then

$$b_{m-1} \ge b_m + c_{m-1} - c_m + \lambda + 2 \ge \beta,$$

[T85, induced-quadrangle inequality; K, Theorem 2.6; LLM strict-case substitution]

whereas $b_{m-1} = \beta + 1 - \psi_{m-1} \le \beta$. Equality would force $\psi_{m-1} = 1$, but then $c_m = \tau_m \psi_{m-1} \le m$, again impossible. Hence

$$c_m \ge m + 2. \tag{31}$$

[T85, induced-quadrangle inequality; K, Theorem 2.6 and Lemma 2.17; LLM strict equality analysis]

If $t = m = d$, then $c_d = m\psi_{d-1}$, so either $c_d = m$ or $c_d \ge 2m$. [K, Lemma 2.17; LLM integrality]

Lemma 7.2 (Exact finite envelope)**.** *For $3 \le d \le 52$, the hypotheses of Proposition 7.1 imply*

$$k \sum_{i=0}^{d} k_i u_i^2 > n$$

[Biggs71, Section 4, multiplicity denominator; K, Theorem 2.10; LLM finite exact certificate at the larger $\varepsilon = 2/(d^3 - 1)$]

unless $c_t = t = m = d$. [LLM]

Proof. All quantities below are rational functions of the integers (d, m, t, r) and, in the refined rows, (c_t, β). The accompanying checker evaluates them as reduced GMP rationals. It enumerates parameter values, not graphs or intersection arrays. [LLM/editorial]

First suppose $t = 2$. Put

$$L_2 = \frac{2m\gamma}{m-1}(1 + \gamma), \qquad q = \frac{m\varepsilon}{1 - m\varepsilon},$$

[LLM/definition]

and

$$Q_2 = \begin{cases} \varepsilon/(1 - 3\varepsilon), & (d, m) = (3, 3), \\ \sum_{j=1}^{d-2} q^j, & \text{otherwise.} \end{cases}$$

[LLM/definition]

The sphere recurrence and (23) give

$$\frac{k_0 + k_1 + \sum_{i=3}^{d} k_i}{k_2} \le L_2 + Q_2.$$

[K, Section 2.2 sphere recurrence and Lemma 2.17; LLM]

Since $u_1 \ge u_*$ and $c_2 = 2$,

$$\frac{kk_1u_1^2}{n} \ge \frac{\frac{2m}{m-1}u_*^2}{1 + L_2 + Q_2}. \tag{32}$$

[LLM]

Now assume $t \ge 3$. Retain $q = m\varepsilon/(1 - m\varepsilon)$. Put $\chi = m(t-1)\gamma$ in the equality case $c_t = t - 1$, and $\chi = m\varepsilon$ otherwise. Define

$$Y_1 = \frac{B}{m}, \qquad Y_2 = \frac{B + m\gamma Y_1/(1 - Y_1)}{m - 1}, \tag{33}$$

[LLM/definition]

and, for $2 \le i \le t-2$,

$$Y_{i+1} = \frac{B + h_iY_i/(1 - Y_i)}{(r + t - 1 - i)(1 - \chi/(i+1))}, \qquad h_2 = 2m\gamma, \quad h_i = \chi\ (i \ge 3). \tag{34}$$

[LLM/definition]

The recurrence (27), strict τ-growth, and $1/\beta = m/k < m\gamma$ imply

$$0 \le y_i \le Y_i < 1 \qquad (2 \le i \le t-1).$$

[K, Lemma 4.2; LLM envelope induction]

The denominator estimate uses $\tau_{i+1} \ge i+1$ only for $i \le t-2$, exactly the range supplied by Kivva's lemma. [K, Lemma 4.2; LLM edge-condition check] Therefore

$$u_{t-1} \ge U := u_* \prod_{i=2}^{t-1}(1 - Y_i).$$

[LLM]

Define

$$\ell_1 = \gamma, \quad \ell_2 = \frac{2m\gamma}{m-1}, \quad \ell_i = \frac{\chi}{(r+t-i)(1-\chi/i)}\ (3 \le i < t), \quad \ell_t = \frac{\chi}{r(1-\chi)}, \tag{35}$$

[LLM/definition]

and

$$L = \sum_{j=1}^{t}\prod_{i=j}^{t}\ell_i, \qquad Q = \sum_{j=1}^{d-t} q^j.$$

[LLM/definition]

For the final ratio, only $\psi_{t-1} \le c_t$ is used; no lower bound on τ_t is assumed. [K, Lemma 2.17; LLM correction relative to the exploratory draft: uses $\psi_{t-1} \le c_t$ only] The sphere recurrence gives $n/k_t \le 1 + L + Q$, and $b_{t-1} \le r\beta$ gives

$$\frac{kk_{t-1}u_{t-1}^2}{n} \ge M := \frac{mc_0U^2}{r(1 + L + Q)}, \tag{36}$$

[LLM]

where

$$c_0 = \begin{cases} t-1, & c_t = t-1, \\ t, & t < m,\ c_t \ge t, \\ m+2, & t = m < d, \\ 2m, & t = m = d,\ c_t \ne m. \end{cases}$$

[LLM, using (29)–(31)]

An exact scan of every permitted tuple with $3 \le d \le 52$ proves that (32) or (36) is greater than 1, except for the following nineteen rows:

case	exceptional parameters
$t = 2$	$(d, m) = (3, 3)$
$c_t = t - 1$	$(d, m, t, r) = (5, 5, 4, 2), (6, 5, 4, 2), (6, 6, 5, 2)$
generic	$(3, 3, 3, 1), (4, 4, 4, 1), (5, 4, 4, 1), (6, 5, 5, 1), (6, 6, 5, 1), (7, 6, 6, 1)$; and $(d, d, d-1, 1)$ for $7 \le d \le 15$

[LLM exact rational audit; verify_unit_coefficient.cpp] Every Y_i in the complete coarse scan is positive with $1 - Y_i > 0.388$. [LLM exact rational audit]

For completeness, we now state the certificates that strengthen the coarse one-sphere bound (36) on those rows. Their published baseline is the one-term estimate in Kivva Proposition 4.6; the new feature is exact parameter retention and, when indicated, an additional nonnegative Biggs-denominator term. Fix the minimal admissible value $c = c_t$ in the relevant row and let β_0 be the least integer satisfying

$$m\beta_0 > d^3, \qquad c < \varepsilon m \beta_0. \tag{37}$$

[LLM/definition]

For fixed $\beta \geq \beta_0$, define $\widehat{Y}_1 = B/m$,

$$\widehat{Y}_2 = \frac{B + \widehat{Y}_1/[\beta(1-\widehat{Y}_1)]}{m-1},$$

[LLM/definition]

and for $2 \leq i \leq t-2$,

$$\widehat{Y}_{i+1} = \frac{B + (c_i^*/\beta)\widehat{Y}_i/(1-\widehat{Y}_i)}{(r+t-1-i)\big(1 + 1/\beta - c/((i+1)\beta)\big)}, \qquad c_2^* = 2, \quad c_i^* = c \ (i \geq 3). \tag{38}$$

[LLM/definition]

Put

$$\widehat{U}_j = u_* \prod_{i=2}^{j} (1 - \widehat{Y}_i).$$

[LLM/definition]

For $t \geq 3$, define

$$\widehat{\ell}_1 = \frac{1}{m\beta}, \quad \widehat{\ell}_2 = \frac{2}{(m-1)\beta}, \quad \widehat{\ell}_i = \frac{c}{(r+t-i)(\beta+1-c/i)} \ (3 \leq i < t), \quad \widehat{\ell}_t = \frac{c}{r(\beta+1-c)}. \tag{39}$$

[LLM/definition]

The last denominator follows from $\psi_{t-1} \leq c_t = c$. [K, Lemma 2.17; LLM correction relative to the exploratory draft: replaces the unsupported $\tau_t \geq t$ step by the source-valid bound $\psi_{t-1} \leq c_t$] Let

$$\widehat{L}_\beta = \sum_{j=1}^{t} \prod_{i=j}^{t} \widehat{\ell}_i, \qquad q_\beta = \frac{\varepsilon m \beta}{\beta + 1 - \varepsilon m \beta}, \qquad \widehat{Q}_\beta = \sum_{j=1}^{d-t} q_\beta^j.$$

[LLM/definition]

The corresponding exact denominator is $D_\beta = 1 + \widehat{L}_\beta + \widehat{Q}_\beta$. [LLM] The bounds above follow directly from

$$c_i \leq c, \qquad \psi_i \leq \frac{c_{i+1}}{\tau_{i+1}} \leq \frac{c}{i+1}, \qquad m - \tau_i \geq r + t - 1 - i,$$

[K, Lemmas 2.17 and 4.2; LLM]

together with the standard-sequence and sphere recurrences. Thus every $\widehat{Y}_i$ is an upper bound for y_i, every $\widehat{\ell}_i$ is an upper bound for k_{i-1}/k_i, and D_β is an upper bound for n/k_t. [LLM]

The principal refined contribution is

$$\mathcal{M}_{t-1}(\beta) = \frac{mc}{r} \frac{\widehat{U}_{t-1}^2}{D_\beta}. \tag{40}$$

[Biggs71, Section 4; K, Theorem 2.10 and Proposition 4.6, one-term $k_{t-1}u_{t-1}^2$ baseline; LLM exact one-sphere refinement]

When needed, the checker also retains

$$\mathcal{M}_t(\beta) = \frac{m\beta \widehat{U}_t^2}{D_\beta}, \qquad \mathcal{M}_{t-2}(\beta) = \frac{mc(t-1)}{r(m-t+2)\beta} \frac{\widehat{U}_{t-2}^2}{D_\beta}. \tag{41}$$

[Biggs71, Section 4; K, Theorem 2.10; LLM strengthens the one-term Kivva Proposition 4.6 certificate by retaining the nonnegative t- or $(t-2)$-sphere summand]

Here $\widehat{U}_t$ is obtained from (27) using the lower bound $b_{t-1} \geq r(\beta + 1 - c)$. The coefficient in $\mathcal{M}_{t-2}$ uses $c_{t-1} \geq t-1$, $b_{t-2} \leq (m-t+2)\beta$, and $b_{t-1} \leq r\beta$. [K, Lemmas 2.17 and 4.2; Biggs71, Section 4; LLM exact recurrence used to retain an extra Biggs-denominator term]

For fixed c, the $\widehat{Y}_i$ decrease and the $\widehat{U}_i$ increase with β; here $c \geq t-1 \geq i+1$ makes every denominator in (38) nondecreasing. Also $\widehat{L}_\beta$ decreases, while q_β increases to $q_\infty = m\varepsilon/(1-m\varepsilon)$. Thus a certificate evaluated at a cutoff $\beta = B_0$ with $\widehat{Q}_\beta$ replaced by $\sum q_\infty^j$ is valid for every $\beta \geq B_0$. [LLM monotonicity] The exact certificates are summarized below; "one" uses (40), "t" adds $\mathcal{M}_t$, and "bridge" checks $\mathcal{M}_{t-1} + \mathcal{M}_{t-2}$ for $\beta_0 \leq \beta < B_0$ and then uses the one-term cutoff certificate.

row	(d, m, t, r, c)	β_0	certificate	cutoff
$t2$	$(3, 3, 2, 2, 2)$	10	t	–
$ep3$	$(3, 3, 3, 1, 6)$	27	bridge	41
$ep4$	$(4, 4, 4, 1, 8)$	64	one	–
$int54$	$(5, 4, 4, 1, 6)$	94	one	–
$eq554$	$(5, 5, 4, 2, 3)$	38	t	–
$eq654$	$(6, 5, 4, 2, 3)$	65	one	–
$int655$	$(6, 5, 5, 1, 7)$	151	one	–
$gen665$	$(6, 6, 5, 1, 5)$	90	one	–
$eq665$	$(6, 6, 5, 2, 4)$	72	one	–
$int766$	$(7, 6, 6, 1, 8)$	229	one	–
$gen776$	$(7, 7, 6, 1, 6)$	147	bridge	180
$gen887$	$(8, 8, 7, 1, 7)$	224	bridge	306
$gen998$	$(9, 9, 8, 1, 8)$	324	bridge	456
$gen10109$	$(10, 10, 9, 1, 9)$	450	bridge	619
$gen111110$	$(11, 11, 10, 1, 10)$	605	bridge	787
$gen121211$	$(12, 12, 11, 1, 11)$	792	bridge	958
$gen131312$	$(13, 13, 12, 1, 12)$	1014	bridge	1130
$gen141413$	$(14, 14, 13, 1, 13)$	1274	bridge	1303
$gen151514$	$(15, 15, 14, 1, 14)$	1575	one	–

[LLM exact rational audit] The smallest exact certified value is

$$1.0000005977102229\ldots > 1,$$

[LLM decimal display of exact positive rational; `verify_unit_coefficient.out`]

and occurs in the cutoff certificate for $gen121211$. [LLM exact rational audit] For each generic row, the checker also verifies that every larger admissible c_t is already covered by the coarse envelope; at $t = m = d$, the next value is $c_t = 3m$ because $c_d = m\psi_{d-1}$. [K, Lemma 2.17; LLM exact audit] Hence all nineteen rows are covered, proving the lemma. [LLM] □

We now treat $d \geq 53$ without parameter enumeration. Put

$$A = \frac{1 + (5m/3 - 1)\varepsilon}{1 - m\varepsilon}, \qquad \delta = A - 1 = \frac{(8m - 3)\varepsilon}{3(1 - m\varepsilon)}. \tag{42}$$

[LLM/definition]

Then $m\varepsilon < 1/50$ and $A < 6/5$. [LLM] We claim

$$0 \leq y_{i+1} \leq \frac{A}{m - \tau_i} \qquad (1 \leq i \leq t - 2).$$

[LLM]

For the base step, $y_1 \leq B/m$ and (27) give

$$y_2 \leq \frac{B + (1/\beta)(B/m)/(1 - B/m)}{m - 1} < \frac{B + m\gamma B/(m - B)}{m - 1} < \frac{A}{m - 1}.$$

[LLM]

Here $B < 51/50 < m - B$, $\gamma < \varepsilon$, and

$$A - B = \frac{m\varepsilon(B + 2/3)}{1 - m\varepsilon} > m\varepsilon.$$

[LLM]

For the induction step $2 \leq i \leq t - 2$, strict growth and the preceding bound give

$$m - \tau_{i-1} \geq r + t - i \geq 3, \qquad 0 \leq y_i \leq A/3 < 2/5,$$

[K, Lemma 4.2; LLM]

so $y_i/(1 - y_i) < 2/3$. Also $\psi_i < \varepsilon k$ and (23) gives $b_i \geq (m - \tau_i)\beta(1 - m\varepsilon)$; substitution in (27) gives the claimed bound. [K, Lemma 2.17; LLM] Telescoping yields

$$u_{t-1} \geq u_1 \frac{r}{r + t - 2} \left(1 - \delta H_{t-2}\right), \tag{43}$$

[LLM]

where $H_j = \sum_{i=1}^{j} 1/i$ and $H_0 = 0$. Every left- and right-tail sphere ratio is at most $q = m\varepsilon/(1 - m\varepsilon)$, apart from the smaller first ratio $1/k < \gamma < q$. [K, Lemma 2.17 and sphere recurrence; LLM] Hence

$$k_t \geq (1 - 2m\varepsilon)n. \tag{44}$$

[LLM]

Combining (28), (43), (44), and $b_{t-1} \le r\beta$ gives

$$k_{t-1}u_{t-1}^2 \ge \frac{n}{k}FR, \tag{45}$$

[LLM]

where

$$F = (1-2m\varepsilon)(1-\varepsilon)^2(1-\delta H_{t-2})^2, \qquad R = \frac{c_t r(m-1)^2}{m(r+t-2)^2}. \tag{46}$$

[LLM/definition]

Outside the endpoint $c_t = t = m = d$,

$$R \ge \frac{m+1}{m}. \tag{47}$$

[LLM]

For $c_t = t-1$, use (30). If $c_t \ge t$ and $t < m$, the function $r/(r+t-2)^2$ has no interior minimum, and its two endpoint values give the result. If $t = m < d$, use (31); if $t = m = d$ but $c_t \ne m$, use $c_t \ge 2m$. [LLM case analysis]

It remains to prove

$$F > \frac{m}{m+1}. \tag{48}$$

[LLM]

Define

$$\Delta_d = 2d\varepsilon + 2\varepsilon + \frac{2(8d-3)\varepsilon}{3(1-d\varepsilon)}H_{d-2}. \tag{49}$$

[LLM/definition]

The exact checker proves

$$\Delta_d < \frac{1}{d+1} \qquad (53 \le d \le 63).$$

[LLM exact rational audit]

The smallest margin occurs at $d = 53$ and equals

$$\frac{246529895422703298250594 9}{1414255080125556482809135 14000} > 0.$$

[LLM exact rational certificate]

For $d \ge 64$,

$$H_{d-2} \le \frac{3}{5}\sqrt{d}. \tag{50}$$

[LLM]

The base case is $H_{62} < 24/5$; the exact margin is

$$\frac{17262497921202896432747309}{19704448068380371125189360 0} > 0.$$

[LLM exact rational certificate]

For the induction step, $10\sqrt{d+1} \le 3(d-1)$ follows from $9d^2 - 118d - 91 > 0$, and therefore

$$\frac{1}{d-1} \le \frac{3}{5}(\sqrt{d+1} - \sqrt{d}).$$

[LLM]

Also

$$\varepsilon < \frac{65}{32d^3}, \qquad d\varepsilon < \frac{1}{2000}.$$

[LLM]

Consequently

$$\begin{aligned} d\Delta_d &< \frac{65}{16d} + \frac{65}{16d^2} + \frac{13}{2\sqrt{d}}\frac{2000}{1999} \\ &\le \frac{65}{1024} + \frac{65}{65536} + \frac{1625}{1999} < \frac{64}{65} \le \frac{d}{d+1}. \end{aligned}$$

[LLM]

The penultimate strict margin is

$$\frac{913198321}{8515420160} > 0.$$

[LLM exact rational certificate]

Thus (49) is less than $1/(d+1)$ for every $d \ge 53$. [LLM]

Applying $\prod(1-x_i) \ge 1 - \sum x_i$ to the five factors in F gives

$$F \ge 1 - (2m\varepsilon + 2\varepsilon + 2\delta H_{t-2}) \ge 1 - \Delta_d > \frac{d}{d+1} \ge \frac{m}{m+1}.$$

[LLM]

This proves (48). Outside the endpoint, (45), (47), and (48) give

$$k_{t-1}u_{t-1}^2 > \frac{n}{k}.$$

[LLM]

Together with Lemma 7.2, this conclusion holds for every $d \geq 3$. [LLM]

Biggs' multiplicity formula is

$$f_1 = \frac{n}{\sum_{i=0}^{d} k_i u_i^2}.$$

[Biggs71, Section 4, theorem on pp. 19–20; K, Theorem 2.10, distance-regular standard-sequence form]

The strict denominator estimate above therefore gives $f_1 < k$. [Biggs71; K, Theorem 2.10; LLM substitution of the certified denominator bound] Since (22) gives $\theta > 0$, Terwilliger's local-eigenvalue theorem puts

$$-1 - \frac{b_1}{\theta + 1} < -1$$

[T86, main theorem; K, Theorem 4.1, exact local-eigenvalue form; LLM checks $f_1 < k$ and $\theta > 0$]

in every neighborhood graph. This contradicts the fact that a disjoint union of cliques has no eigenvalue below -1. [K, Lemma 2.20; LLM] Therefore

$$c_t = t = m = d. \tag{51}$$

[LLM]

At the endpoint, (25), $\tau_d = m = d$, and $\tau_i \leq d - 1$ for $i < d$ force

$$\tau_i = i \qquad (1 \leq i \leq d).$$

[K, Lemma 4.2 and Corollary 4.3; LLM integer-chain argument]

Then $c_d = d = \tau_d \psi_{d-1}$ gives $\psi_{d-1} = 1$. If some $\psi_j \geq 2$, choose j maximal and put $i = j + 1$. Monotonicity gives

$$i + 1 = c_{i+1} = \tau_{i+1}\psi_i \geq c_i = \tau_i \psi_{i-1} \geq 2i,$$

[PS, Section 2.1; K, Lemma 2.17; LLM]

impossible. Thus $\psi_i = 1$ for every i, and

$$c_i = i, \qquad b_i = (d - i)\frac{k}{d}.$$

[K, Lemma 2.17; LLM endpoint substitution]

Since $k/d = b_{d-1} \in \mathbb{Z}$, these are the Hamming intersection numbers. [LLM] Egawa's classification gives a Hamming or Doob graph. [Ega81, main classification, pp. 108–125; K, Theorem 2.25 (journal)/2.24 (arXiv), exact restatement] A Doob graph has $k = 3d$, contrary to $k > d^3$, so X is Hamming. [K, proof of Theorem 4.7; LLM] □

8. Completion of the proof

Proof of Theorem 1.1. Suppose a nonidentity automorphism has support density $\rho < \gamma$. Proposition 4.1 makes X Delsarte-geometric with $m \leq d$. Lemma 5.1 rules out a transition, and Lemma 5.2 gives (20). [LLM]

If $\mu \geq 3$, Proposition 6.1 makes X Johnson. If $\mu = 2$, Proposition 7.1 makes X Hamming. If $\mu = 1$, Proposition 6.2 contradicts $\rho < \gamma$. These cases exhaust the positive integer μ. Therefore every graph outside the Johnson and Hamming families satisfies

$$\mathrm{motion}(X) \geq \gamma n = \frac{n}{d^3}.$$

[LLM]

□

9. Dependency and audit ledger

The proof imports the following published statements. [LLM/editorial]

(1) From Pyber–Skresanov [1]: the primitive-relation and distinguishing-number bounds (Propositions 2.10 and 2.12), the geodesic-load argument (Proposition 2.8), Babai's spectral motion bound (Proposition 2.13), $2\lambda \leq k + \mu$ (Lemma 2.4), and Bang–Koolen geometricity (Proposition 2.5). [PS, cited items]

(2) From Kivva [3]: Metsch's theorem and the clique-geometry criterion (Theorem 2.12 and Proposition 2.13 in the arXiv numbering), the Delsarte bound and geometric identities (Section 2.3 and Lemmas 2.14, 2.17–2.20 in journal numbering), Biggs' formula (Theorem 2.10), Terwilliger's inequality and local eigenvalue theorem (Theorems 2.6 and 4.1), the Johnson local tools (Theorems 3.1, 3.9, 2.21, Lemma 3.12, Proposition 3.11), strict τ-growth (Lemma 4.2), the dual-graph spectrum and incidence transfer (Lemmas 2.26–2.27, 5.1, 5.5), the $m = 2$, $\mu = 1$ motion result (Proposition 5.13), and the Hamming/Doob endpoint classification. [K, cited items]

(3) From Babai–Wilmes [2]: an equivalent parameterized statement of Metsch's theorem (Theorem 1.2). [BW, Theorem 1.2]

Original-source use map. Unlike the earlier draft, the annotations at the actual points of use now cite the older papers directly: [LLM/editorial responding to source-audit request]

(1) Delsarte [4], Section 3.3.2, is pointed to at the clique bound used to prove $q_v \geq m$; Kivva Section 2.3 supplies the exact distance-regular formulation. [Del73, Section 3.3.2; K, Section 2.3]
(2) Metsch [6, 7], Result 2.2 (1995) and Result 2.1 (1999), is pointed to at (13) and (18); the modern exact restatements are Koolen–Bang Proposition 5.1 and Kivva Theorem 2.12. [Met95, Result 2.2; Met99, Result 2.1; KB10, Proposition 5.1; K, Theorem 2.12]
(3) Koolen–Bang [5], Theorem 5.3, is invoked directly with $M = \lceil m \rceil$ in the branch $m \leq d-1$; Pyber–Skresanov Proposition 2.5 is its convenient reformulation. [KB10, Theorem 5.3; PS, Proposition 2.5]
(4) Biggs [8], Section 4, theorem on pp. 19–20, is cited both at the multiplicity formula and at the refined t- and $(t-2)$-sphere certificates; Kivva Theorem 2.10 is the standard-sequence restatement. [Biggs71, Section 4; K, Theorem 2.10]
(5) Terwilliger [9], the induced-quadrangle intersection-number inequalities, is cited wherever Kivva Theorem 2.6 is used; Terwilliger [10], the main theorem, is cited at the local-eigenvalue contradiction restated as Kivva Theorem 4.1. [T85; K, Theorem 2.6; T86; K, Theorem 4.1]
(6) Egawa [11], the classification on pp. 108–125, is cited at the Hamming/Doob endpoint, alongside Kivva Theorem 2.24/2.25. [Ega81; K, Theorem 2.24 (arXiv)/2.25 (journal)]

Kivva numbering concordance. The arXiv and journal versions differ by one in several places: arXiv Lemmas 2.16–2.19 correspond to journal Lemmas 2.17–2.20, and arXiv Theorem 2.24 corresponds to journal Theorem 2.25. The exact Metsch statement is Theorem 2.12 in the arXiv version. [LLM/editorial, checked against both versions]

Every row below records a hypothesis check rather than an additional theorem. [LLM/editorial]

Interface	**Hypotheses checked here**
PS standing assumption	The valency-2 cycle case is dispatched after Theorem 1.1.
PS primitive-relation bound	Every nontrivial relation in a primitive distance scheme is connected and has diameter at most d.
Metsch	Applied with integer parameter $s = d$; both strict inequalities appear in (17).
Koolen–Bang	Used only in the branch $m \leq d-1$, with $M = \lceil m \rceil \leq d-1$, $-M \leq -m < 1-M$, and $M^2\mu < \lambda$ all displayed before Theorem 5.3 is invoked.
Clique-geometry criterion	In the branch $d-1 < m \leq d$, integer rounding gives exactly d lines through every vertex and $k > d^3 \geq d^2$, so Kivva Proposition 2.13 (arXiv) applies.
Kivva local tools	The local graph is connected; $b^+ < 13/12$; every interlacing contradiction is strict; $k > d^3 \geq m^3$.
Kivva dual spectrum	$k > d^3 \geq m^2$; both the positive and negative sides of the dual zero-weight spectral radius are bounded.
Strict τ growth	$\mu = 2$, $c_t < \varepsilon k$, and $\varepsilon < 1/d^2 \leq 1/m^2$; no conclusion about τ_t is used in the refined endpoint denominator.
Finite exact envelope	Every integer (d, m, t, r) with $3 \leq d \leq 52$ is scanned; all nineteen coarse exceptions are covered by the displayed rational certificates.
Terwilliger local eigenvalue	Biggs' formula gives the strict inequality $f_1 < k$, and $\theta > 0$.
Hamming endpoint	The surviving case is exactly $c_t = t = m = d$; $k/d = b_{d-1}$ is integral; Doob valency $3d$ is excluded.

Remark 9.1 (Barrier of the present uniform structural split). The integer clique-count completion of the Metsch reduction removes the former coefficient-3/2 bottleneck and makes coefficient 1 possible. For a hypothetical support threshold $\gamma = a/d^3$, however, the branch $m \leq d-1$ would still require

$$a\big(d(d-1)^2 + 1\big) < d^3$$

[LLM]

from the Bang–Koolen inequality. The right-hand ratio tends to 1 as $d \to \infty$. Thus a uniform coefficient strictly larger than 1 requires another idea for the band of smallest eigenvalues just below d, not merely tighter scalar estimates in the $\mu = 2$ branch. [LLM/heuristic]

Remark 9.2. The exponent 3 remains natural for this exact-geometricity strategy: small support gives $\mu \lesssim \rho k$, while $m = O(d)$ and $\lambda \gtrsim k/d$ make the Bang–Koolen condition $m^2\mu < \lambda$ operate at scale $\rho = O(d^{-3})$. [LLM/heuristic]

References

[1] L. Pyber and S. V. Skresanov, *On the automorphism group of a distance-regular graph*, J. Combin. Theory Ser. B **172** (2025), 94–114; arXiv:2312.00383.

[2] L. Babai and J. Wilmes, *Asymptotic Delsarte cliques in distance-regular graphs*, J. Algebraic Combin. **43** (2016), 771–782; arXiv:1503.02746.

[3] B. Kivva, *A characterization of Johnson and Hamming graphs and proof of Babai's conjecture*, J. Combin. Theory Ser. B **151** (2021), 339–374; arXiv:1912.11427.

[4] P. Delsarte, *An algebraic approach to the association schemes of coding theory*, Philips Res. Rep. Suppl. **10** (1973), vi+97.

[5] J. H. Koolen and S. Bang, *On distance-regular graphs with smallest eigenvalue at least* $-m$, J. Combin. Theory Ser. B **100** (2010), 573–584.

[6] K. Metsch, *A characterization of Grassmann graphs*, European J. Combin. **16** (1995), 639–644.

[7] K. Metsch, *On a characterization of bilinear forms graphs*, European J. Combin. **20** (1999), 293–306.

[8] N. L. Biggs, *Intersection matrices for graphs*, in *Combinatorial Mathematics and its Applications* (Oxford, 1969), Academic Press, London, 1971, 15–23.

[9] P. Terwilliger, *Distance-regular graphs with girth* 3 *or* 4. *I*, J. Combin. Theory Ser. B **39** (1985), 265–281.

[10] P. Terwilliger, *A new feasibility condition for distance-regular graphs*, Discrete Math. **61** (1986), 311–315.

[11] Y. Egawa, *Characterization of* $H(n, q)$ *by the parameters*, J. Combin. Theory Ser. A **31** (1981), 108–125.